Daniil Vlasenko

✉ vlasenko.daniil.vl@gmail.com
🌐 daniilvlasenko.info

Research Interests: Neuroscience, Network Science,
Data Analysis, Mathematical Modeling

Education

HSE University, Institute for Cognitive Neurosciences
PhD, Doctoral School of Cognitive Science
Moscow, Russia, November 2025 – present

HSE University, Institute for Cognitive Neurosciences
MS, Cognitive Sciences and Technologies
Moscow, Russia, September 2023 – June 2025

Saint Petersburg State University, School of Mathematics and Mechanics, Department of Statistical Modelling
BS, Applied Mathematics and Computer Science
St. Petersburg, Russia, September 2019 – June 2023

Additional Education

CNeuro, Theoretical and Computational Neuroscience Summer School
Beijing, China, 7–14 July 2026

Neuromatch Academy, NeuroAI course
Online, 14–25 July 2025

Neuromatch Academy, Computational Neuroscience course
Online, 8–26 July 2024

Mediterranean School of Complex Networks
Grado, Italy, 30 June – 5 July 2024

Bioinformatics Institute, Professional retraining program "Algorithmic Bioinformatics"
St. Petersburg, Russia, September 2022 – January 2023

Selected Publications

Vlasenko, D., Saranskaia, I., & Zakharov, D. (2026). Hypergraphs from multivariate connectivity: caCoh-based EEG/MEG representation. arXiv.

(Q1 Scientific Journal) Krivonosov, M., Nazarenko, T., Ushakov, V., Vlasenko, D., Zakharov, D., Chen, S., Blyus, O., & Zaikin, A. (2025). Analysis of Multidimensional Clinical and Physiological Data with Synolitical Graph Neural Networks. Technologies, 13(1), 13.

Additional publications can be found on my Google Scholar page.

Selected Conferences

Volga Neuroscience Meeting 2025, The 7th International Conference “Neurotechnologies and Neurointerfaces”, topic “From Pairwise to High-Order: Hypergraph Methods and Multivariate Connectivity Metrics for EEG/MEG” (poster presentation)
Nizhny Novgorod, Russia, 25–29 August 2025

International School and Conference on Network Science: **NetSci 2025**, topic “Ensemble-Based Graph Representation of fMRI Data for Cognitive Brain State Classification” (poster presentation)
Maastricht, the Netherlands, 2–6 June 2025

Baltic Forum 2024: Neuroscience, Artificial Intelligence and Complex Systems, VIII Scientific School “Dynamics of Complex Networks and their Applications”, topic “*Ensemble methods for representation of fMRI, EEG/MEG data in graph form for classification of brain states*” (poster presentation).
Kaliningrad, Russia, 19–21 September 2024

Scholarships and Fellowships

Russian National PhD Research Scholarship, the program supporting PhD students and their dissertation research.
August 2026 – present

Combined Master's-PhD track at HSE University, the talent program designed for students enrolled on full-tuition scholarships.
September 2023 – present

Academic Development Program (New Scientist category) at HSE University, the talent management program aimed at supporting the professional development of promising teachers and researchers.
January 2024 – May 2025

Honors and Awards

Winner (1st Place), Information Technology & Mathematics Section, 10th All-Russian Youth Scientific Forum “Science of the Future – Science of the Youth”.
September 2025.

Certificate of Excellence “**Best Diploma of the Year 2025**”, HSE University Institute for Cognitive Neuroscience.
June 2025

(Selected) Research experience

Agence Nationale de la Recherche project “DyndriteSymphony: Dynamics of nonlinear dendritic computation in cortical pyramidal neurons and interneurons”
External collaborator
September 2025 – present

My tasks: developing mathematical models of synaptic clustering that explain experimental observations, and investigating how passive dendrites can implement nonlinear computations.

Russian Science Foundation Interdisciplinary Grant 24-68-00030 "Next Generation Cognitive Artificial Intelligence"
Research-assistant
June 2024 – present

My tasks: developing graph and hypergraph representations of fMRI and EEG/MEG data, and applying network analysis and network-based machine learning to brain-state classification.

Teaching Experience

Supervision of course and diploma works of undergraduate students, **HSE University**
September 2024 – present

Seminar instructor (a few classes per course), “Advanced Computational Neuroscience”, “Some Topics of Calculus, Differential Equations and Nonlinear Dynamics”, **HSE University**
September 2024 – present

Languages

Russian (native), English (B2)

Skills

Programming
Python — NumPy, Pandas, Matplotlib, SciPy, Scikit-Learn, PyTorch, PyTorch Geometric,

iGraph, NiLearn, MNE; R — dplyr, tidyr, ggplot2, iGraph; C++; MySQL; algorithms and data structures; Linux-based operating systems; familiar with remote server operation (terminal, Bash), version control systems (git, GitHub), HTML, CSS, JavaScript and Selenium; preparing documents and presentation slides using LaTeX.

Data analysis and machine learning
Classical data analysis; analysis of categorical data; network analysis; machine learning including deep learning; databases (SQL queries). Analysis and preprocessing of fMRI, EEG/MEG data; analysis of neural electrode data (neuropixels probes).

Mathematics
Statistics; probability theory; graph theory; linear algebra; mathematical analysis; (partial) differential equations; analytic geometry; optimization methods.

Additional Interests

Bouldering, visual arts